



LEARN DATASCIENCE IN 2023

COMPLETE ROADMAP

5 MONTHS

ZERO MONEY NEEDED

All Free



- Introduction to Python for Data Science
 - Python Introduction
 - Python , Jupyter and Anaconda Installation
 - Wlkthrough of Jupyter and Anaconda
 - Python Syntax and Semantics
 - Python Data Structures
 - Python Functions and Lambdas
 - Python List comprehension and generators
 - Python regular expressions
 - Python File IO and Resource Operations
- Advanced Python for Data Science
 - Numpy deepdive
 - Pandas deepdive
 - Matplotlib & Seaborn – Visualization
 - Common Modules and Packages

Corey Shafer	https://www.youtube.com/watch?v=YYXdXT2l-Gg&list=PL-osiE80TeTt2d9bfVyTiXJA-UTHn6WwU
MLWithAP	https://www.youtube.com/watch?v=Kk-ob0bQKhg&list=PL47eid5T-F5S1U_xiTfFR3f8myu7YGF2I

- Mathematics for Machine Learning (2 weeks)
 - Statistics – Descriptive and Inferential
 - Linear Algebra
 - Calculus & Probability

Khan Academy - <https://www.khanacademy.org/math/statistics-probability>

Khan Academy - <https://www.khanacademy.org/math/calculus-1>

Organic Chemistry Tutor -

<https://www.youtube.com/@TheOrganicChemistryTutor/playlists>

- Exploratory Data Analysis
 - Data Gathering and Extraction
 - SQL (MySQL). NoSQL (Mongo)
 - CSV and JSON files
 - Web Crawling / Scraping / APIs / Beautiful Soup
 - Cloud - S3
 - Unstructured data (Video, Speech , Text)
 - Data Preprocessing & Analysis
 - Univariate, Bivariate and Multivariate Analysis
 - Outliers and Anomalies detection
 - Data Cleansing - Null value, Imputations, Duplicate treatment

- Dispersion and Distribution
- Data Split
 - Train, Test, Validation set
 - Sampling technique and Stratification Strategy
 - Bias and Variance trade off
 - Handling Bias and Imbalance

Coursera / IBM - <https://www.coursera.org/learn/ibm-exploratory-data-analysis-for-machine-learning>

- Introduction to Machine Learning
 - Machine learning Landscape
 - Machine learning end to end project lifecycle
 - Regression Vs Classification
 - Supervised Vs Unsupervised
 - Batch Vs Online
 - Linear Regression
 - Multi Variable Linear Regression
 - Logistic Regression
 - Scikit learn and Statsmodel
- Feature Engineering and Feature Selection
 - Feature Engineering
 - New feature creation
 - Variable transformation
 - Feature Encoding
 - Handling Categorical and Numerical features
 - Binning

- Feature Selection
 - PCA
 - Dimensionality Reduction techniques
 - Multicollinearity
 - Forward/Backward/Stepwise selection
 - Lasso
 - Filter/Wrapper/Embedded
- Feature Scaling
 - Standardization
 - Normalization

Coursera Andrew Ng's -

<https://www.coursera.org/specializations/machine-learning-introduction>

Josh Starmer

<https://www.youtube.com/@statquest>

Machine Learning - 2

5 Weeks

- Advanced Supervised Learning
 - Naïve Bayes Classifier
 - k-NN Classifier
 - Support Vector Machines (Regressor and Classifier)
 - Ensemble Techniques
 - Decision Tree
 - Bagging
 - Random Forest
 - Boosting
- Model Selection and Tuning

- Hyper Parameter Tuning
- Model Performance measures
- Bias and Variance
- Overfitting vs Underfitting
- Cross validation
- GridSearchCV Vs RandomizedSearchCV
- Regularization – L1 and L2
- Pipelining

- Unsupervised Learning
 - K-means clustering
 - KNN
 - Hierarchical Clustering
 - Anomaly detection
 - Dimensionality Reduction Techniques / PCA
 - SVD
 - DBSCAN

- Production Deployment.
 - Deployment scenario and strategies
 - ML Pipeline
 - Flask and Heroku
 - Introduction to FastAPI
 - Deployment to AWS ECS
 - Monitoring and Continuous performance measure
 - MLOps

Stanford CS229 –

https://www.youtube.com/watch?v=jGwO_UgTS7I&list=PLoROMvodv4rMiGQp3WXShTMGgzqpfVfbU

Kaggle :

<https://www.kaggle.com/learn>

Deep Learning

5 Weeks

- Neural Network and Deep Learning Fundamentals
 - Perceptron
 - Activation and loss function
 - Gradient Descent
 - Batch Normalization
 - Introduction to TensorFlow and Keras
 - Transfer learning and regularization
- Computer Vision (2 Weeks)
 - CNN
 - Convolution, Pooling and Padding
 - CNN architectures and ImageNet Challenge
 - Object Detection
 - Semantic Segmentation
- Natural Language Processing

- RNNs
- Tokenization, Stemming and Lemmatization
- LSTMs and GRUs
- Time Series analysis
- Advance Language Models – BERT, GPT3
- Attention is all you need

- Autoencoders and GANs
 - Generative Network and Adversarial Network
 - Variational Autoencoders
 - Convolution and DCGAN
 - Application of GANs

- Reinforcement Learning
 - RL framework
 - Markov Chain
 - Policy Gradient Methods
 - Type of RL systems
 - Q Learning

Andrew Ng's CS230 Stanford –

https://www.youtube.com/watch?v=PySo_6S4ZAg&list=PLoROMvodv4rOABXSygHTsbvUz4G_YQhOb

Coursera / Deeplearning.AI

<https://www.coursera.org/specializations/deep-learning>

MIT 6S.191

https://www.youtube.com/watch?v=7sB052Pz0sQ&list=PLtBw6njQRU-rwp5_7C0oIVt26ZgjG9NI

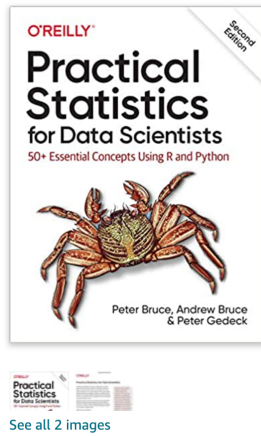
Stanford CS231n

<https://www.youtube.com/watch?v=vT1JzLTH4G4&list=PLC1qU-LWwrF64f4QKQT-Vg5Wr4qEE1Zxk&index=2>

Google's Machine Learning Course –

<https://developers.google.com/machine-learning/crash-course>

Books :



Practical Statistics for Data Scientists, 2e: 50+ Essential Concepts Using R and Python Paperback – 29 June 2020

by Peter Bruce (Author), Andrew Bruce (Author), Peter Gedeck (Author)

★★★★★ 763 ratings

[See all formats and editions](#)

Kindle Edition
₹2,634.93

Read with Our **Free App**

Paperback
₹3,941.00 ✓prime

12 New from ₹3,941.00

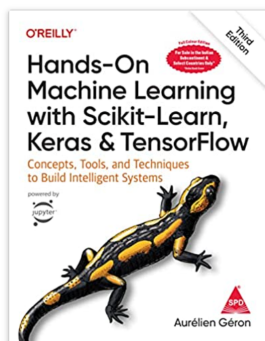
EMI starts at ₹188. No Cost EMI available [EMI options](#)

Save Extra with 4 offers

Cashback: ₹200 cashback & ₹1,800 welcome rewards on [Amazon Pay ICICI Credit Card](#) . 5% cashback every time on shopping. Apply now! T&C. Not applicable on Amazon Business Transactions | [Details](#)

No Cost EMI: Avail No Cost EMI on select cards for orders above ₹3000 | [Details](#)

✓ [See 2 more](#)



Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, Third Edition (Full Colour Print) Paperback – 10 October 2022

by Aurélien Géron (Author)

★★★★★ 4 ratings

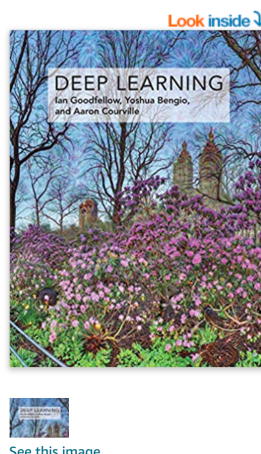
[See all formats and editions](#)

Paperback
₹3,000.00 ✓prime

2 New from ₹3,000.00

EMI starts at ₹143. No Cost EMI available [EMI options](#)

★★★★★ 4 ratings



Deep Learning (Adaptive Computation and Machine Learning series) Hardcover – 18 November 2016

by Aaron Courville (Author), Ian Goodfellow (Author), Yoshua Bengio (Author)

★★★★★ 1,917 ratings

Part of: Adaptive Computation and Machine Learning series (13 books)

[See all formats and editions](#)

Kindle Edition
₹4,346.25

Read with Our **Free App**

Hardcover
₹5,500.00

8 New from ₹5,490.00

Paperback
from ₹3,000.00

2 Used from ₹3,000.00

1 New from ₹4,000.00

EMI starts at ₹263. No Cost EMI available [EMI options](#)

Save Extra with 4 offers

Cashback: ₹200 cashback & ₹1,800 welcome rewards on [Amazon Pay ICICI Credit Card](#) . 5% cashback every time on shopping. Apply now! T&C. Not applicable on Amazon Business Transactions | [Details](#)

No Cost EMI: Avail No Cost EMI on select cards for orders above ₹3000 | [Details](#)

Podcasts:

Lex Friedman

<https://www.youtube.com/c/lexfridman>

Adrej Karpathy – Not a Podcast – but a Youtube Channel

<https://www.youtube.com/@AndrejKarpathy>

Other things :

1. Build few (3+) end to end project portfolio and showcase it on LinkedIn
 - a. Don't build Titanic / Iris data set. Do something new and novel to stand out
 - b. Do it end to end with deployment on cloud.
2. Build an Online presence on LinkedIn by posting and engaging with other ML community members.
 - a. LinkedIn is new resume. Spend time and polish it.
3. Practice, Practice and Practice.
4. Learn from Others – Especially Kaggle notebooks and how they approach any problem.